
Origin and Dependency Chain for EML Placement

A PREPRINT

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Claim

The EML operator

$$\text{eml}(x, y) = e^x - \ln y$$

is a continuous-shadow composite inside Phase Calculus, not a primitive. The operation layer it uses descends from the lifted-state stack. This note records the placement spine that the restored main paper reconnects to the already-operational native selector evolution.

Dependency chain

Work	Date / DOI	Load-bearing role
CF000	Mar. 18, 2026 / 10.5281/zenodo.19098778	Primitive distinguishability, non-discharge, same-axis saturation, orthogonal re-articulation, proto- $\mathbb{N}$, algebra unlock.
CF00	Mar. 8, 2026 / 10.5281/zenodo.19099916	Completion chain $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$, complex field $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$, topology, differentiability, carrier, quotient derivative, QGT.
Farey Remainder Recursion	Mar. 24, 2026 / 10.5281/zenodo.19210954	Exact remainder/refinement conjugacy and finite-resolution logarithmic stopping.
CF19	Mar. 27, 2026 / 10.5281/zenodo.19267927	Exact lifted object $\widehat{\Xi} = (A, q, \theta, \kappa, c)$, visible/state distinction, exact return, anchor (55, 89).
Phase Calculus Formalisation	Apr. 2026	Operator core $\{Q, B, L\}$, selector macro-calculus, quotient criterion $\Pi \circ E = G \circ \Pi$, continuous shadow theorem.
Operational Utility	Apr. 18, 2026	Native π spigot: certified streaming, bank-indexed access, scalar-collapse obstruction.
This EML paper	Apr. 22, 2026	EML as continuous-shadow composite after primitive operation descent.

Compact equation

$$\begin{aligned} &\text{survivor mark} \rightarrow \text{primitive roll} \rightarrow i, \theta, \pi \\ &\rightarrow \mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \\ &\rightarrow (q, c) \rightarrow \{Q, B, L\} \rightarrow \Pi \circ E = G \circ \Pi \\ &\rightarrow \text{Sub}_{PC}, \text{Exp}_{PC}, \text{Log}_{PC} \rightarrow \text{eml}_{PC} . \end{aligned}$$

Result

$$\text{eml}_{PC}(x, y) = \text{Sub}_{PC}(\text{Exp}_{PC}(x), \text{Log}_{PC}(y)) = e^x - \ln y.$$

EML compresses continuous expression syntax. Phase Calculus supplies the discrete lifted-state operation source beneath that syntax.